

Maxwell Costa
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Skills

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- Schematic capture, PCB layout, circuit design, debug and bring-up
 - Intermediate in Python, Arduino, Lua, Java, Fusion 360, KiCad
 - Basic in AutoCAD
 - Basic in MS Word, Google Docs, and Google Sheets

Extracurricular Activities

MSEF Regional & State fairs, August 2026 - April 2026

- Awards: 2nd Place (regional fair), 3rd place (state fair), NOAA "Taking the Pulse of the Planet" Special Award (Regional Fair).
- Researched and implemented a new algae clearing process using electroflotation without damaging the algae, successfully reducing algal turbidity by 90-100%
- Created a modular prototype with custom 3D printed housing and a custom designed PCB integrating an ATmega328PB microcontroller.
- Provisional Patent Pending.

Amherst Regional High School Lemelson-MIT InvenTeam, October 2023 - July 2025

- One of two primary electrical design engineers developing a vest to monitor the health data of emergency response workers [Project Article](#).
- Designed a custom PCB using KiCad to connect sensors and power to the Arduino and the Blues wireless notecard in a compact profile.
- Hand-soldered and built PCBs, including surface mount and through-hole components.
- Developed firmware to run on an Arduino Nano 33 BLE to interface with a maxrefdes117 heart rate and SPO2 sensor and a thermistor.
- Developed firmware to transmit data over LTE networks using a Blues Wireless Notecard Carrier module.
- Developed backend Python software to receive and process data from the health monitor Web Link.
- Supported team presentation of our health monitoring device at the 2023 LemelsonMIT EurekaFest.
- Patent pending US20250375158A1.

Amherst Regional High School Robotics Team (FTC), October 2023 - present

- Work on the hardware team to construct our robot's general frame and develop mechanisms using CAD.
- Current software lead, developing both manual and autonomous code.
- Mentor new members on robot software and firmware.
- General electrical, hardware, and software development and debugging.
- Volunteered at community outreach events to introduce more people to our robotics team and STEM pathways.

Amherst Regional High School Computer Science Club, February 2025 - present

<https://arhs.arps.org/about/arhs-weather-data>

- Team member of weather station installation on school grounds to monitor weather and send data to the Citizen Weather Observation Program.
- Recurring presentations at the Arbors senior home on cybersecurity and technology literacy.

University of Massachusetts Amherst Turing Program (July 2024)

- Selected to participate in a summer learning program on computer science fundamentals.
- Introduced to notable topics such as machine learning, cryptography, and information networking.
- Completed exercises in Python to learn topics noted above such as a Caesar cipher.

Education

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- Amherst Regional High School (Expected Graduation June 2027)
 - Electives: AP Computer Science (A, 5), Electrical Engineering (A+), Mechanical Engineering (A), Robotics (A+), Computer-Aided Design (A+), and Advanced Engineering and Design (A+)
 - Innovation Career Pathways program that focuses on introducing students to STEM careers

Athletics

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- ARHS Cross Country, 2026
 - ARHS Ultimate Frisbee, 2025
 - ARPS/ARHS Ski team, 2022 - 2025